

Bottom versus Charm Jet Tagging using Machine Learning

Nishank Gite,^{1,*} Johannes Wagner,^{1,2,†} and Marjorie Shapiro^{1,2,‡}

¹*Department of Physics, University of California, Berkeley, CA 94720, USA*

²*Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA*

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The process of jet tagging, especially the distinction between bottom (b) and charm (c) jets, is essential in high-energy physics experiments, notably those conducted at the Large Hadron Collider (LHC). This research employs machine learning methodologies, particularly Graph Neural Networks (GNNs), to improve the precision and efficiency of b versus c jet tagging, with an emphasis on enhancing performance in the low transverse momentum (p_T) region, where jets tend to overlap significantly. We adapted a pre-trained GNN model, which was initially optimized on a diverse dataset, by utilizing specialized datasets derived from $W + \text{jets}$ and $t\bar{t}$ events. The evaluation of the model's performance, through metrics such as Receiver Operating Characteristic (ROC) curves and p_T versus efficiency plots, indicated advancements in the discrimination between c and b jets, particularly when trained on $W + \text{jets}$ data. However, challenges were encountered in effectively rejecting light jets within the low p_T range. Future research will aim to address c versus $\bar{c}$ classification through secondary vertex charge reconstruction, alongside further investigation into advanced model architectures, data augmentation strategies, and hyperparameter tuning to create more robust and accurate jet tagging models suitable for intricate particle physics contexts.

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I. INTRODUCTION

Jet tagging has been a critical task in high-energy physics for decades, particularly in experiments conducted at the Large Hadron Collider (LHC). The ability to accurately distinguish between different types of jets, such as those originating from bottom (b) quarks and charm (c) quarks, is essential for various analyses.

Accurate bottom versus charm jet tagging has several significant benefits. For instance, it is crucial for studying

the couplings of the Higgs boson to charm and bottom quarks. The Higgs to charm coupling ($H \rightarrow c\bar{c}$) is an important but challenging measurement due to the overwhelming background from other processes, and the ability to effectively distinguish charm jets can significantly improve the sensitivity of these measurements. Similarly, precise b-jet tagging is essential for studying the Higgs to bottom coupling ($H \rightarrow b\bar{b}$), which is the dominant decay mode of the Higgs boson and plays a key role in verifying the Standard Model predictions. Beyond Higgs physics, optimal b vs. c tagging can enhance the study of heavy flavor production, top quark decays, and the search for exotic particles that preferentially decay to heavy quarks. [1]

This study investigates the use of machine learning to improve the accuracy and efficiency of bottom versus charm jet tagging. Specifically, we utilize Salt, a multi-modal framework for creating graph neural network taggers in high-energy physics, which is already a strong and well-optimized model. [2] Our hypothesis is that by adding auxiliary tasks and optimizing the dataset via additional features, we can potentially improve performance. We specifically aim to extend these improvements to the low p_T regime (transverse momentum). Enhancing performance in this regime is particularly important because many new physics scenarios predict the production of particles with relatively low p_T jets (e.g. SUSY and Dark Matter Candidates), and low p_T jets are often more challenging to tag due to their lower energy and more substantial overlap with background processes, thus improving tagging performance here can provide significant benefits. [3]

* nishankgite@berkeley.edu

† johannes_wagner@berkeley.edu

‡ mdshapiro@lbl.gov

II. JETS

Jet tagging is one step in a longer end-to-end analysis process in high-energy physics. The process begins with a collision, such as proton-proton (pp), and involves several stages to simulate and analyze the resulting particle interactions. Initially, the collision event is simulated using a program like Pythia, which models the particle shower process. This includes the fragmentation and hadronization of the primary particles. Following this, the GEANT software is employed to simulate the interaction of these particles with the detector, providing a realistic picture of what the detector would actually record.

Jets are collimated sprays of particles originating from a single high-energy particle, such as a quark or gluon. To determine the particles that originate from a common source, the data must be clustered using jet clustering algorithms. This involves grouping together particles that are close in phase space. The most commonly used algorithm is the anti- k_T jet clustering algorithm, defined by the metric:

$$d_{ij} = \min(p_{T,i}^{2p}, p_{T,j}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \quad d_{iB} = p_{T,i}^{2p},$$

where $p_{T,i}$ is the transverse momentum of particle i , $\Delta R_{ij} = \sqrt{(\Delta y_{ij})^2 + (\Delta \phi_{ij})^2}$ is the distance between particles i and j in the rapidity-azimuth plane, R is a parameter controlling the jet size, and $p = -1$ for the anti- k_T algorithm. [4]

Jet tagging is the final step of this full process, where we determine the origin particle for each of the jets that have been clustered. This comprehensive process results in the identification of jets and their respective parent particles.

In the specific case of b vs. c tagging, the challenge is heightened due to the difficulty in distinguishing between the different decays. After a collision and the production of b or c quarks, gluon radiation occurs, leading to a cascade of quark formation via processes like $g \rightarrow q\bar{q}$. [5] Both b and c quarks will hadronize to form B mesons and D mesons, respectively, and the complexity arises because B mesons frequently decay into D mesons. This overlap in the decay products of B and D mesons makes it challenging to distinguish between b-jets and c-jets, and is the reason why b vs. c tagging is so difficult to optimize.

III. EXPERIMENTAL STRATEGY

A. Machine Learning Techniques for Jet Tagging

Jet tagging plays an essential role in the field of particle physics, particularly in the identification and categorization of various jet types during high-energy physics experiments, such as those performed at the Large Hadron

Collider (LHC). The application of machine learning techniques, particularly those based on graph theory, has gained prominence in addressing this challenge, owing to the intricate, non-Euclidean data structures that arise from particle detector outputs. The inherent graphical characteristics of the data facilitate the utilization of graph neural networks (GNNs), which are adept at discerning relationships among entities within a non-linear spatial framework. [6] Figure 1 presents differ-

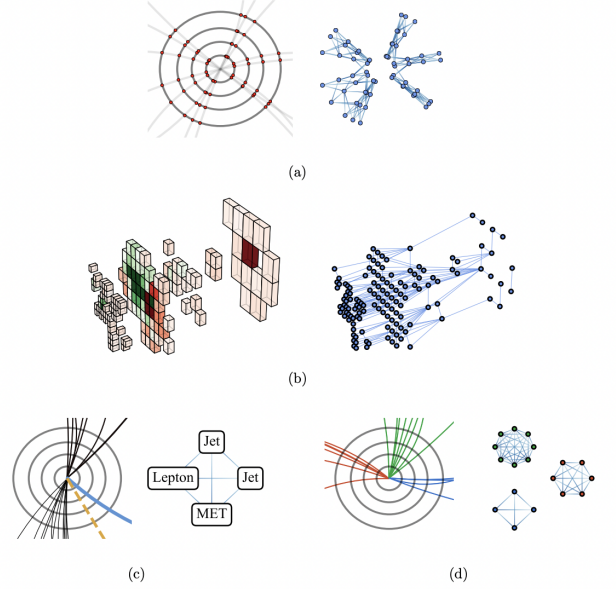


FIG. 1. Particle Physics detector data can be well represented as a graph for many applications: (a) track clustering based off detector hits, (b) calorimeter segmentation, (c) multiple event classification, and (d) jet classification. [7].

ent representations of particle physics detector data as graphs, underscoring the adaptability of graph-based machine learning methodologies across various phases of data analysis. These graph-based representations allow machine learning models to better leverage the complex spatial and relational structure of particle physics data, which leads to improvements in jet tagging accuracy and efficiency. [8]

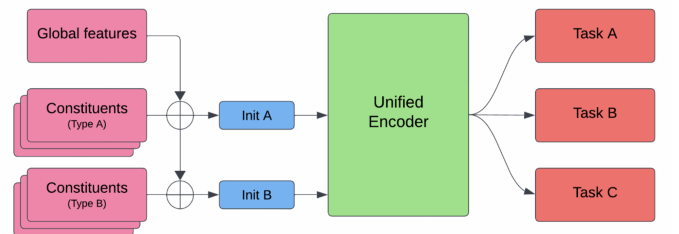


FIG. 2. This is the generic model information flow architecture for Salt: Multimodal Multitask Machine Learning for High Energy Physics [2].

Figure 2 illustrates the framework of a standard model employed in jet tagging which exemplifies the unified encoder methodology, wherein global features, such as characteristics of jets, and constituent features, including track or calorimeter data, are initially processed through distinct initialization networks prior to their integration within a unified encoder. This design allows the model to effectively manage various modalities of input data, which is essential for multitask learning in high-energy physics. By facilitating the model’s ability to execute multiple tasks—such as classification, regression, and segmentation—via implementation of auxiliary tasks in addition to our primary jet classification, this strategy enhances the overall efficacy and versatility of jet tagging models.

B. Data

In our analysis, we utilized a pre-trained Graph Neural Network (GNN) that was initially trained on a broad dataset. This dataset included $t\bar{t}$ events, which provided statistics in the low p_T regime, and ZZ' events, covering the high p_T range. While this pre-trained model offers a solid foundation for general-purpose jet tagging, our objective was to enhance performance specifically in the low p_T regime. To achieve this, we focused our analysis on two targeted datasets: W +jets and $t\bar{t}$ events, allowing us to fine-tune the model for better performance at low p_T .

In $t\bar{t}$ events, both b and c jets are commonly produced. A typical $t\bar{t}$ decay involves one top quark decaying into a W boson and a b quark ($t \rightarrow Wb$). The b quark hadronizes into a B meson, which can subsequently decay into a D meson and then further into lighter particles like kaons and pions, ultimately leading to final-state particles that are detected in the experiment. Additionally, the W boson from the top quark decay can decay hadronically into two quarks ($W \rightarrow q\bar{q}'$), where one or both quarks could be c or b quarks. These c quarks hadronize into D mesons, which follow similar decay chains to detectable particles. The overlap between b and c jets poses a challenge, making differentiation between the two more complex. In W +jets processes, the W boson is produced alongside additional jets. While these jets are frequently light-flavor jets, they can also originate from gluons or heavy-flavor quarks, leading to the formation of b or c jets. By concentrating on these decay pathways within the W +jets and $t\bar{t}$ datasets, our goal is to improve the GNN’s ability to distinguish between b and c jets, particularly in the low p_T regime, where such differentiation is more challenging.

The features used in our jet tagging model range from basic kinematic variables to detailed track-level information, all contributing to improved tagging performance. Key features include the transverse and longitudinal impact parameters (d_0 , $z_0\sin\theta$), which help identify secondary vertices associated with heavy-flavor jets by

telling you the distance of closest approach of the track to the beamline perpendicular to the beam axis and parallel respectively. The significance of these impact parameters (IP3D_signed_d0_significance: the ratio of the transverse impact parameter and its associated uncertainty ($\frac{d_0}{\sigma_{d_0}}$), IP3D_signed_z0_significance: the ratio of the longitudinal impact parameter and its associated uncertainty ($\frac{z_0}{\sigma_{z_0}}$)) further enhances the model’s ability to distinguish between primary and secondary vertices. Kinematic features, such as the corrected transverse momentum and pseudorapidity ($pt_btagJes$, $eta_btagJes$), provide crucial information about jet momentum and spatial distribution. Additionally, hit-based features ($numberOfPixelHits$, $numberOfSCTHits$) reflect track quality and contribute to accurate jet reconstruction. Emphasizing these features ensures the model effectively distinguishes jet types while maintaining high accuracy.

C. Performance Metrics

Receiver Operating Characteristic (ROC) curves are commonly used to evaluate the performance of classification models, particularly in jet tagging. The ROC curve represents the trade-off between true positive rate (efficiency) and false positive rate (background rejection) at various decision thresholds. In our graphs, the x-axis represents the jet efficiency, which is the fraction of correctly identified jets of a certain flavor, while the y-axis shows the background rejection, which is the inverse of the false positive rate for background jets, which are commonly light jets. A higher background rejection indicates that the model is more effective at distinguishing jets, and the model’s performance varies for different jet types across different efficiencies. These curves are crucial in assessing how well the model can separate jets from background across a range of operating points.

IV. GN2 MODEL PERFORMANCE

In this section, we evaluate the performance of the GN2 model by comparing the pre-trained model with the new model trained on W +jets and $t\bar{t}$ data. The performance is assessed using several metrics, including ROC curves and p_T vs efficiency plots. These metrics help us understand how well the model distinguishes between different jet types, particularly in the low p_T regime, which is our area of focus. The following plots provide insights into the GN2 model’s performance across various scenarios.

Figure 3 shows the ROC curve comparing the pre-trained GN2 model to the newly trained model across the full p_T range. The x-axis represents c-jet efficiency, and the y-axis shows background rejection (the inverse of the false positive rate) for both b-jets and light jets. The two models perform similarly across most of the p_T spectrum, with the new model showing slight improvements in background rejection for c vs b jets (up to 20%

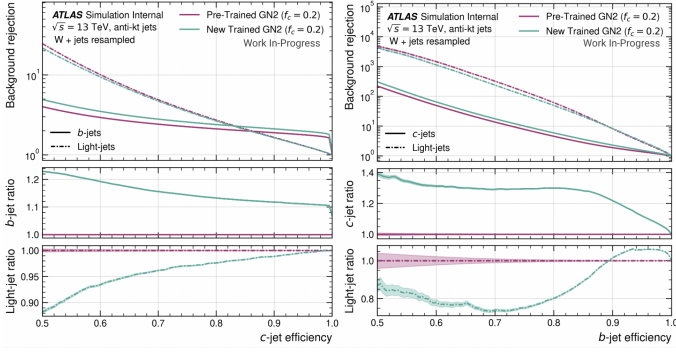


FIG. 3. This is the ROC curve comparing the pre-trained GN2 model to the new $W + jets$ data trained model for all p_T .

for b -jet rejection and 40% for c -jet rejection), but worse performance when compared to light jets (by up to 10% for c jets and 20% for b jets).

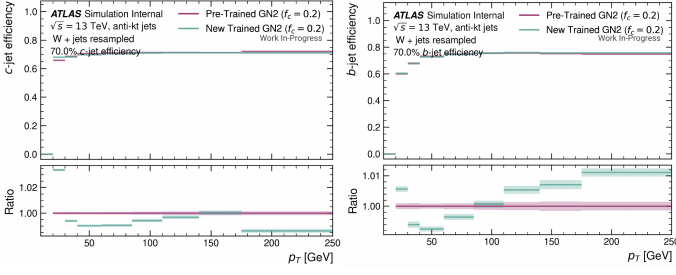


FIG. 4. This is the p_T versus efficiency plots for $W + jets$ data at a 70% working point for the model, so it is trying to maintain 70% efficiency across all p_T .

Figure 4 presents the p_T vs efficiency plots at a 70% working point for both c -jets and b -jets. While the model is tuned globally to achieve 70% efficiency, this plot reveals that efficiency can dip below the pre-trained at low p_T . This dip reflects the inherent challenges of tagging jets in the low p_T regime, where the model struggles more to maintain its performance. Specifically, when we compare the two models, we see that for b -jet efficiency, we have almost identical performance within a 1% margin of error, with slightly worse efficiency for p_T around 50 GeV, but a bump in performance for extremely low p_T . For c -jet efficiency, we see that the performance is almost identical but worse for the new model, except for extremely low p_T , which is ideally what we want.

Figure 5 displays the ROC curve focused on the low p_T region ($p_T \leq 50$ GeV). Here, the model's performance degrades compared to the full p_T range, as seen by the lower background rejection at similar c -jet and b -jet efficiencies. This is expected due to the increased difficulty of distinguishing between jet types at low p_T which we confirmed in Fig. 4, where their properties tend to overlap more. However, the new GN2 model still demonstrates similar relative performance as in Fig. 3, with notable improvements over the pre-trained model for b versus c tagging,

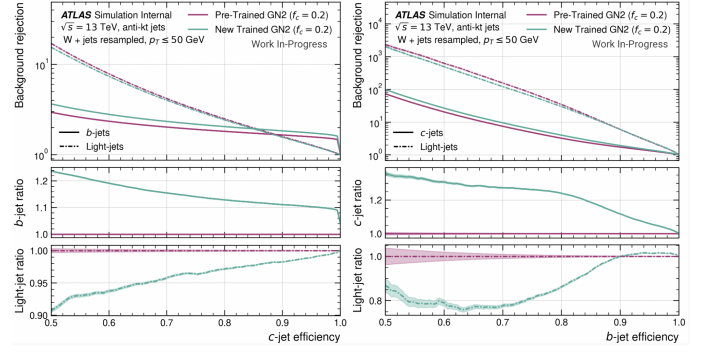


FIG. 5. This is the ROC curve comparing the pre-trained GN2 model to the new $W + jets$ data trained model with a cut for $p_T \leq 50$ GeV.

but significantly worsening light jet rejection up to 10% for the c -jets and 20% for b -jets.

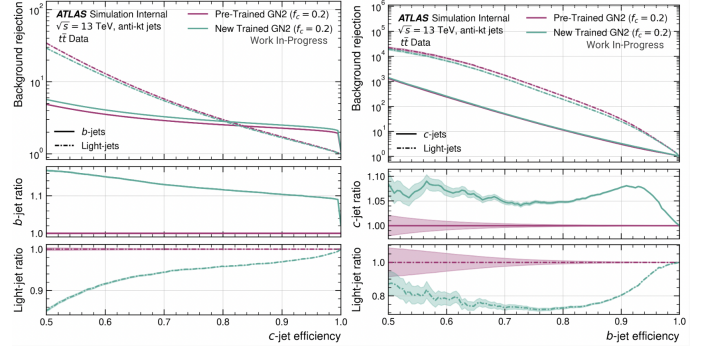


FIG. 6. This is the ROC curve comparing the pre-trained GN2 model to the new $t\bar{t}$ data trained model for all p_T .

Figure 6 shows the ROC curve comparing the pre-trained GN2 model to the newly trained model across the full p_T range. Again, like in Fig. 3, the two models perform similarly across most of the p_T spectrum, with the new model showing slight improvements in background rejection for c vs b jets (up to 20% for b -jet rejection and 10% for c -jet rejection), but worse performance when compared to light jets (by up to 15% for c jets and 30% for b jets). From just this initial glance, it seems like the choice to use $W + jets$ data may result in some performance benefits, but there is a bias since we are training and testing on the same $W + jets$ data.

Figure 7 presents the p_T vs efficiency plots at a 70% working point for both c -jets and b -jets. Again, (similar to Fig. 4), this plot reveals that efficiency can dip below the pre-trained at low p_T . Specifically, when we compare the two models, we see that we have practically identical performance, with variations that are less than 1%. Visually, we can also see they look identical, and this is likely because both models were trained with a $t\bar{t}$ sample, and it also implies that the ZZ' data for higher p_T had very little to no impact on truth model performance.

Finally, figure 8 displays the ROC curve focused on the

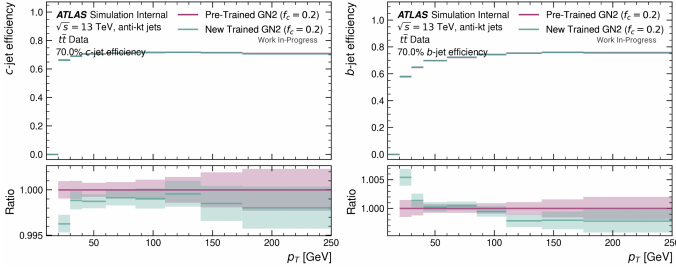


FIG. 7. This is the p_T versus efficiency plots for $t\bar{t}$ data at a 70% working point for the model, so it is trying to maintain 70% efficiency across all p_T .

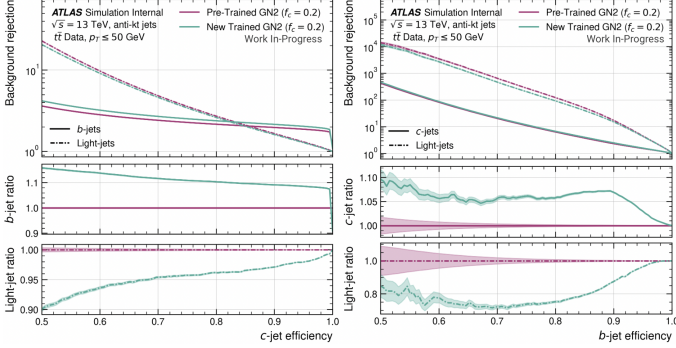


FIG. 8. This is the ROC curve comparing the pre-trained GN2 model to the new $t\bar{t}$ data trained model with a cut for $p_T \leq 50$ GeV.

low p_T region ($p_T \leq 50$ GeV). Here, again the model's performance degrades compared to the full p_T range, as seen by the lower background rejection at similar c -jet and b -jet efficiencies. The important thing to note here is that the rejection ratios did not significantly change from Figure 6, which means that our model still outperforms at low p_T for our specific classification task, though is still worse at rejecting light jet background which is a significant tradeoff.

V. CONCLUSION

This report presents an evaluation of the performance of both a pre-trained Graph Neural Network (GN2) and a newly trained GN2 model across two separate datasets: W + jets and $t\bar{t}$ events. The primary objective was to enhance the model's efficacy in the low p_T regime, a particularly challenging area for jet tagging due to the overlapping characteristics of jets at low transverse momentum.

Utilizing various performance metrics, such as ROC curves and p_T versus efficiency plots, we found that re-training the GN2 model on the W + jets dataset resulted in modest improvements in background rejection for c versus b jets, albeit at the cost of some performance degradation in the rejection of light jets. Similarly, re-training on the $t\bar{t}$ dataset yielded analogous outcomes,

with enhanced discrimination between c and b jets but a decline in the rejection of light jets. Notably, the W + jets dataset appeared to produce a more effective tagger overall compared to the $t\bar{t}$ dataset, indicating that the additional information derived from W + jets processes may play a more significant role in differentiating between jet types in challenging conditions.

The p_T versus efficiency plots at a 70% working point underscored the model's difficulties in sustaining consistent efficiency at low p_T , further emphasizing the intricacies of low p_T jet tagging. Nevertheless, the new GN2 model exhibited robustness across various p_T ranges, and while its performance in the low p_T region showed slight degradation, it still provided considerable improvements over the pre-trained model for specific classification tasks.

In summary, our investigation emphasizes the critical role of selecting suitable training datasets tailored to the specific physics analysis objectives. The newly developed GN2 model, which has been trained on W + jets data, presents a promising avenue for enhancing jet tagging efficacy in the low p_T range. Nevertheless, the balance between improved discrimination of c versus b jets and the concomitant decrease in light jet rejection underscores the necessity for ongoing optimization and refinement of the model, especially in complex regions of phase space. Future research may focus on the implementation of additional data augmentation methods, enhancements to model architecture, or the adoption of specialized training approaches to further improve performance in these challenging contexts.

VI. FUTURE DIRECTIONS

As an extension of this study, we are now focusing on the classification of c vs $\bar{c}$ jets. To tackle this, we are working on a secondary vertex charge reconstruction auxiliary task by leveraging the truth quark charges, which are determined by the PDG ID of the specific particle. We are currently training models optimized for c vs $\bar{c}$ classification and expect to have results in the near future.

In addition to the aforementioned strategies, there are numerous potential pathways for enhancing the tagging of b versus c jets that merit investigation. One promising strategy involves the refinement of the model architecture through the incorporation of sophisticated neural network methodologies, including attention mechanisms and graph-based message-passing networks. Such modifications could facilitate a more nuanced understanding of the complex interactions between tracks and secondary vertices within jets, which are critical for effective jet tagging.

Furthermore, the implementation of data augmentation techniques could serve to artificially expand the training dataset, particularly in the low p_T region. This approach may alleviate the performance decline typically observed at low p_T . Additionally, broadening the

training dataset’s diversity by integrating various physics processes that generate b and c jets could enhance the model’s robustness across a broader spectrum of operational conditions.

Another significant avenue for improvement lies in the optimization of hyperparameters, including learning rates, batch sizes, weight initializations, and network depth. Conducting a comprehensive hyperparameter search could uncover configurations that enhance the performance of b versus c tagging. Moreover, exploring loss functions that impose direct penalties for misclassifications in particularly challenging areas, such as low p_T , may further propel the model towards superior tagging accuracy.

In summary, these prospective directions are poised to advance the development of more precise and resilient jet tagging models, with a particular focus on enhancing performance in difficult scenarios and improving the

differentiation between various jet types.

CODE AVAILABILITY

The code for this paper can be found at <https://github.com/afinitus/bot-vs-charm-jet-tagging/tree/tutorial>.

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